

Edge cases (search comes back empty) — Hints

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Only 2 hints — work through them in order, and don't jump ahead until you've genuinely tried. Each hint has 3 depth levels: **Basic** (the plain idea), **Intermediate** (proper Python), **Advanced** (how a real RAG system handles the messy edge cases). Read Basic first even if you already know Python — it's the fastest way to spot exactly what each deeper level adds.

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Hint 1 — The idea, and the exact pieces

Basic Version

Ask a question that's clearly outside your document set's coverage (like asking a policy document about the weather). Your search will come back with nothing genuinely relevant.

The question this exercise really tests: what does your graph do with "nothing found"? Does it notice and say so honestly, or does it hand the model an empty (or near-empty) block of "context" and let it guess an answer anyway?

Things to use:

- Your search node, saving its results into the state (an empty list if nothing found).
- A new conditional edge, checked right after search, that looks at whether that list is empty.
- A small "can't answer" node that returns a clear, honest message instead of an answer.

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Intermediate Version

This exercise is about adding an explicit check, not a new capability. After your search node runs, check whether it actually found anything worth using — and if not, route to a distinct "can't answer" response instead of letting the reasoning node try anyway with nothing to work from.

```
def has_results(state: dict) -> str:
    if not state.get("sources"):
        return "no_grounding"
    return "reason"
```

The important design point: this check has to happen *after* search runs, as its own routing decision — not folded silently into the reasoning node's prompt, where "I found nothing" and "I found something thin" can blur together into a guess.

The exact pieces:

- **What "nothing useful" means** — an empty list is the obvious case.
- **The new node** — `cant_answer_node` should return something the caller can clearly tell apart from a real, grounded answer — a distinct field, or a clearly worded message, not just a short version of a normal answer.
- **Where this check lives** — a conditional edge right after the search node, checked before the reasoning node ever runs, so the model is never even given the chance to improvise from nothing.

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Advanced Version

An empty list is the easy case to catch. A harder, more common case: search returns *something*, but it's barely related — a vector similarity search almost never returns literally zero results, it returns the *closest* chunks it has, even when none of them are actually relevant. A thin, low-scoring result is just as misleading as no result at all, because your "nothing useful" check won't catch it — the list isn't empty, so the graph sails on to `reason_node` and the model dutifully "answers" from context that doesn't really support an answer.

The real design question isn't "is the list empty" — it's "how relevant does a result actually need to be before it counts as grounding, and what do you do with a result that's borderline?"

Answering that needs a similarity score, which most vector retrievers can give you alongside each chunk, and a threshold you choose deliberately (not an arbitrary guess — start from a few real examples of "this is genuinely relevant" versus "this is a coincidental near-miss," and pick a number that separates them).

The extra pieces needed:

- Your retriever returning a `(chunk, score)` pair per result instead of just the chunk, so the routing check has something to compare against a threshold.
- A `RELEVANCE_THRESHOLD` constant, and a check like `max(score for _, score in results) >= RELEVANCE_THRESHOLD` instead of just `not results`.
- A place to log every "no grounding" and "below threshold" event with the original task text — not to change behavior in this exercise, but because reviewing those logs later is how a real project discovers gaps in its document set (the same questions keep failing to ground → that content is missing, not just phrased badly).

Sketch the threshold check yourself before checking Hint 2.

Difference between Basic, Intermediate, and Advanced: Basic and Intermediate both treat "found nothing" as "the list is empty" — true, but only half the problem. Advanced adds the harder, more common case: a non-empty list that's still not good enough to ground an answer, caught with a relevance-score threshold instead of a length check, plus a logging habit that turns "search fails silently on some questions" into "here's the list of questions search actually fails on, and why" — the difference between noticing a gap once and being able to systematically close it.

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Hint 2 — The plan, and almost the whole thing

Basic Version

```
after search_node runs, state has "sources" (a list, maybe empty)

function has_results(state):
    if sources list is empty:
        return "no_grounding"
    return "reason"

add a new node cant_answer_node:
    return a message like "I don't have grounding for this in my documents"

wire:
    add_conditional_edges(after search_node, has_results, {"no_grounding": cant_a

test:
    ask something outside your document set -> should land on cant_answer_node
```

Here's almost the whole thing — just try running it and reading it line by line:

```
def has_results(state):
    if not state.get("sources"):
        return "no_grounding"
    return "reason"

def cant_answer_node(state):
    return {"answer": "I don't have grounding for this in my documents.", "ground

graph.add_node("cant_answer_node", cant_answer_node)
graph.add_conditional_edges("search_node", has_results, {"no_grounding": "cant_an
```

Expected output if you run just this: nothing on its own — invoke the graph with a fully out-of-scope question (like "what's the weather today?") and check that `result["answer"]` is the honest fallback message, not a guess.

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Intermediate Version

```

define:
  def has_results(state: dict) -> str:
    if not state.get("sources"):
      return "no_grounding"
    return "reason"

  def cant_answer_node(state: dict) -> dict:
    return {"answer": "I don't have grounding for this in my documents.", "gr

register:
  graph.add_conditional_edges(
    "search_node",
    has_results,
    {"no_grounding": "cant_answer_node", "reason": "reason_node"},
  )

test:
  invoke with a question with zero overlap with your document set
  confirm the final state's "answer" is the honest fallback, not a guess

```

```

def has_results(state: dict) -> str:
    sources = state.get("sources", [])
    if not sources:
        return "no_grounding"
    return "reason"

def cant_answer_node(state: dict) -> dict:
    return {
        "answer": "I don't have grounding for this in my documents.",
        "grounded": False,
    }

```

Register `cant_answer_node` with `graph.add_node(...)`, wire the conditional edge, and test both an in-scope and out-of-scope question before comparing against the [Solution](#).

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Advanced Version

```
RELEVANCE_THRESHOLD = some number you chose from real examples

function has_results(state):
    scored_sources = state's sources, each with a similarity score
    if scored_sources is empty OR the best score is below RELEVANCE_THRESHOLD:
        log this task as "no grounding"
        return "no_grounding"
    return "reason"
```

Here's almost the whole thing — fill in the missing piece yourself:

```
import logging

logger = logging.getLogger("grounding")
RELEVANCE_THRESHOLD = 0.75

def has_results(state: dict) -> str:
    scored_sources = state.get("scored_sources", [])

    # your turn: compute the best (highest) score among scored_sources,
    # treating an empty list as "no score at all" -- then return
    # "no_grounding" if scored_sources is empty OR the best score is
    # below RELEVANCE_THRESHOLD, otherwise return "reason". Log the
    # task text either way you go, using logger.info(...).
    ...
```

Fill in the threshold check and the logging call yourself, then compare all 3 of your finished versions against the [Solution](#).

Difference between Basic, Intermediate, and Advanced: Basic and Intermediate both check only whether the list is empty — a real search backend that always returns *some* chunks (just possibly bad ones) will sail right past that check. Advanced replaces "is it empty" with "is the best result actually good enough," using a similarity-score threshold chosen from real examples, and adds a logging habit that turns silent failures into a reviewable record of exactly which questions your document set can't currently answer.

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Full solution: [Show me the solution](#)